ITEC632 - Applied Data Mining

Semester 1 - 2025

Assessment 3

Predictive Data Mining Project

Submitted By

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1. Introduction

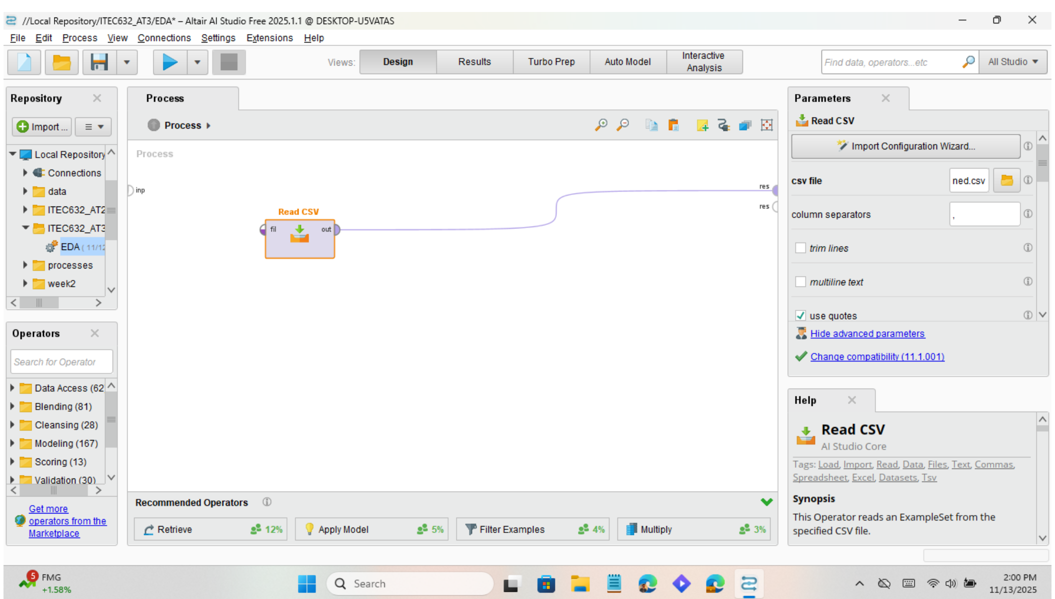
Heart disease remains one of the leading causes of morbidity and mortality worldwide, making early detection and risk assessment a critical public health priority. Advances in machine learning provide opportunities to develop predictive models that can identify individuals at risk based on lifestyle and clinical attributes. This project applies data mining techniques using RapidMiner to evaluate two models, Decision Tree and Logistic Regression, for heart disease prediction. The workflow includes preprocessing categorical variables, applying cross-validation to ensure robust evaluation, and extracting performance metrics such as accuracy, precision, recall, F-measure, AUC, and Kappa. By comparing model outputs and interpreting variable importance, the project aims to determine which approach offers the best balance between predictive performance and interpretability. Beyond technical evaluation, ethical considerations such as fairness, privacy, transparency, accountability, and potential harms are explored to ensure responsible deployment of AI in healthcare. Together, these steps provide a comprehensive framework for assessing both the technical and ethical suitability of machine learning models in clinical screening contexts.

2. Exploratory Data Analysis

a. Load and Inspect the dataset

The original huge dataset could be loaded in RapidMiner, but due to my license being a standard one, I could not process the whole set of data, so I trimmed the data down to 9500 rows.

a. Read CSV



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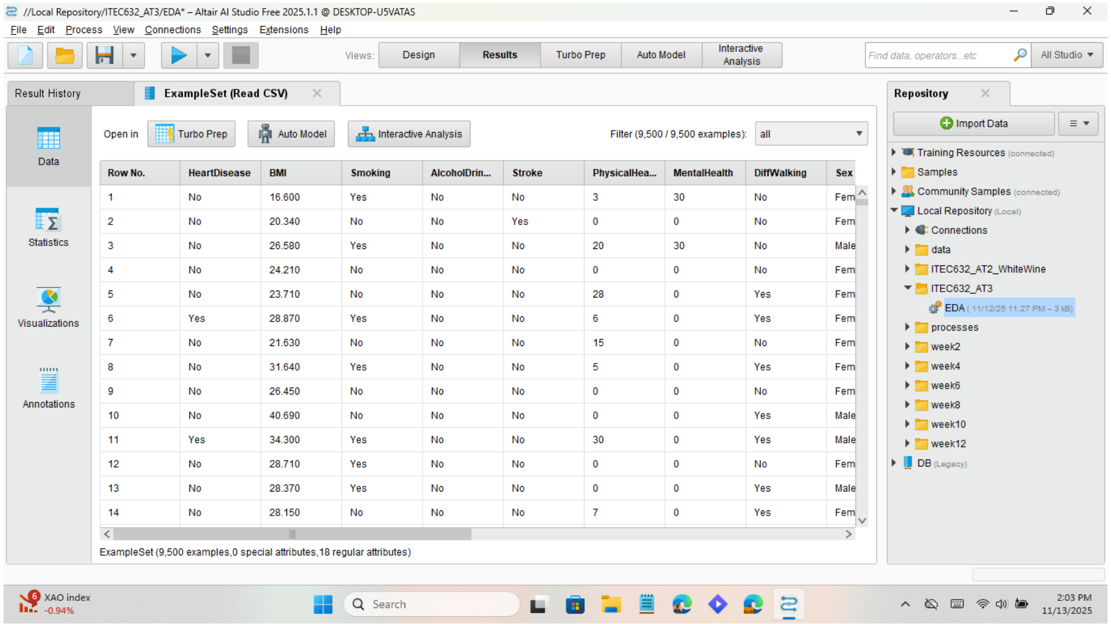
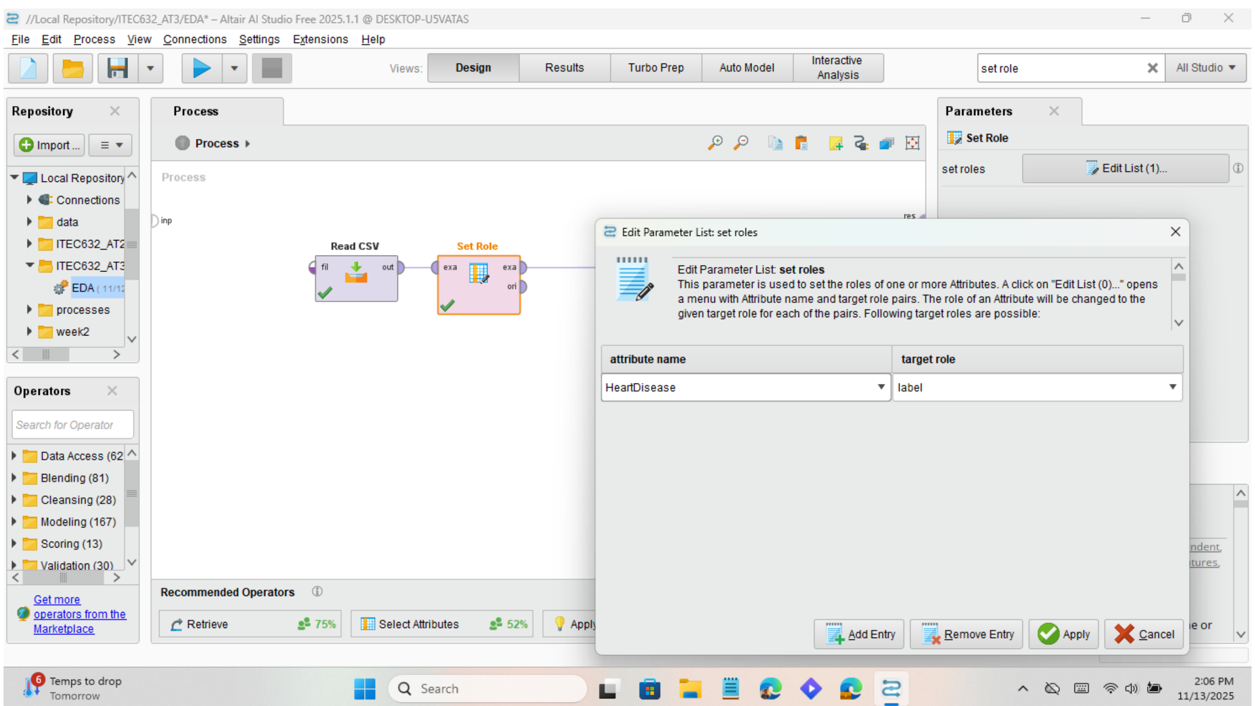


b. Set Role



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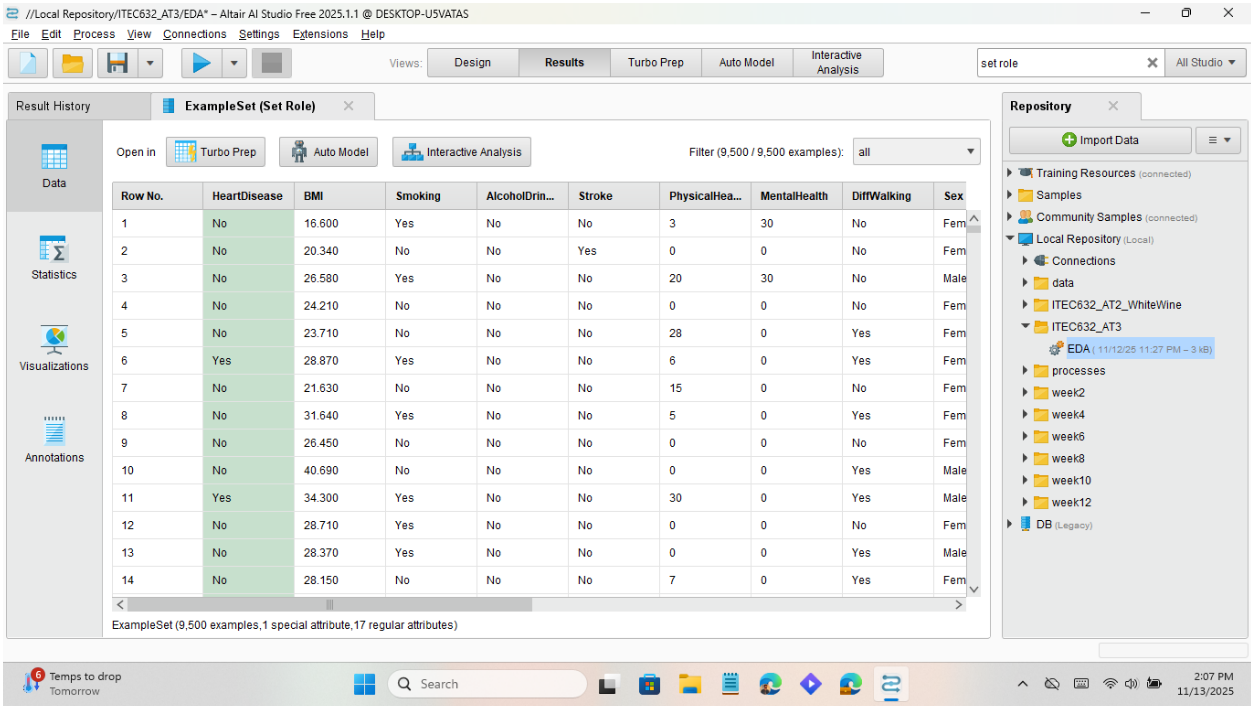
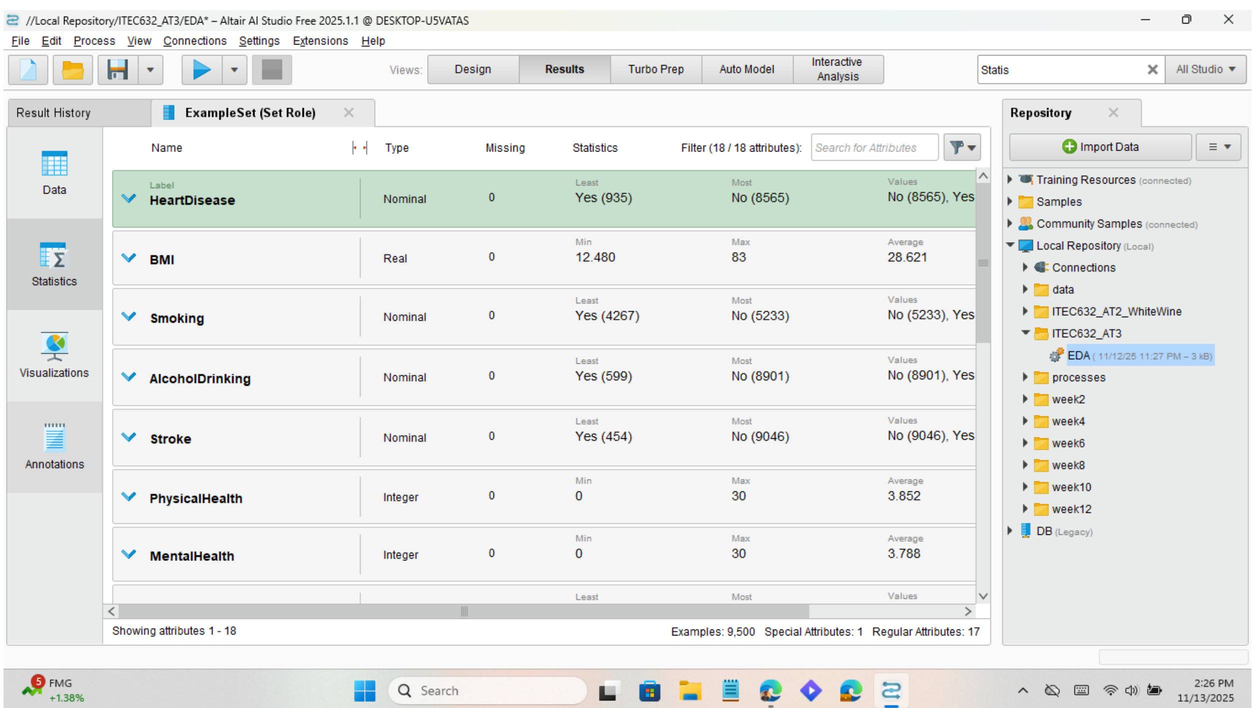
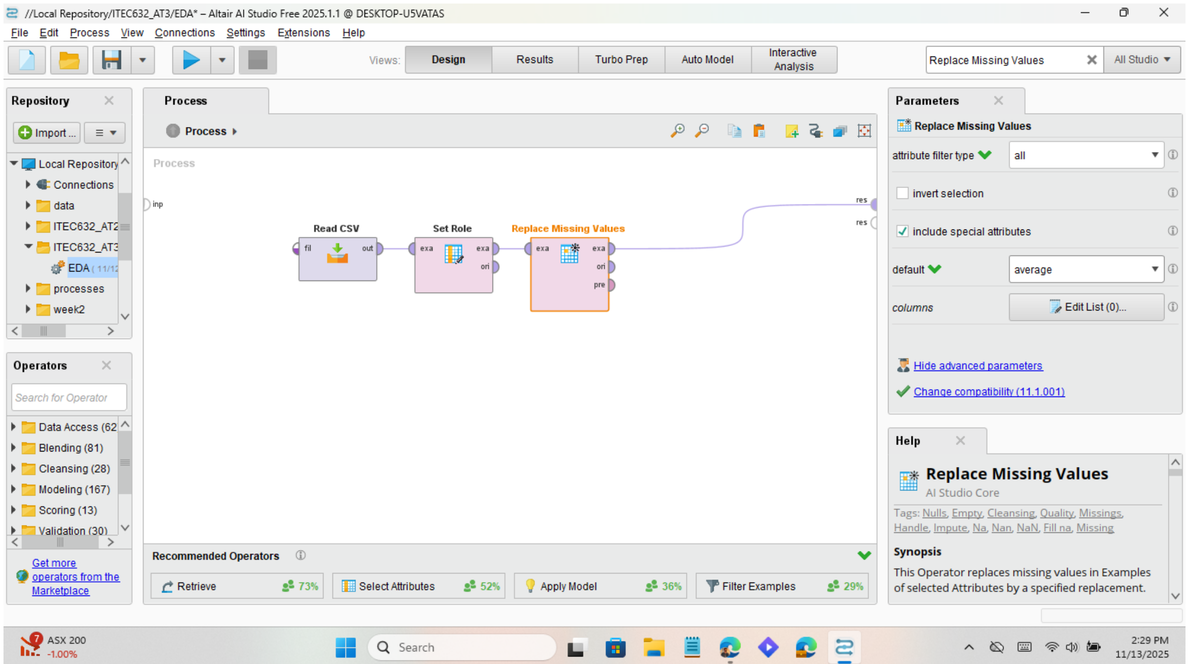


c. Statistics



b. Clean and Prepare

a. Replace Missing Values



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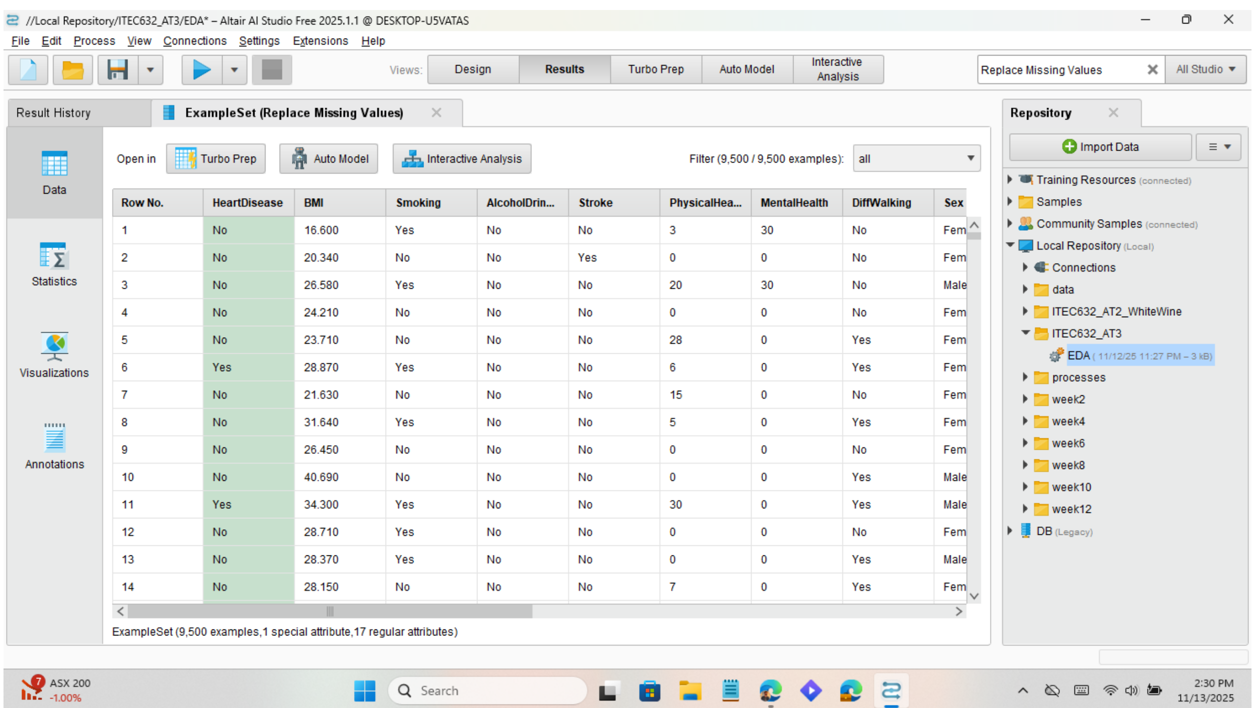
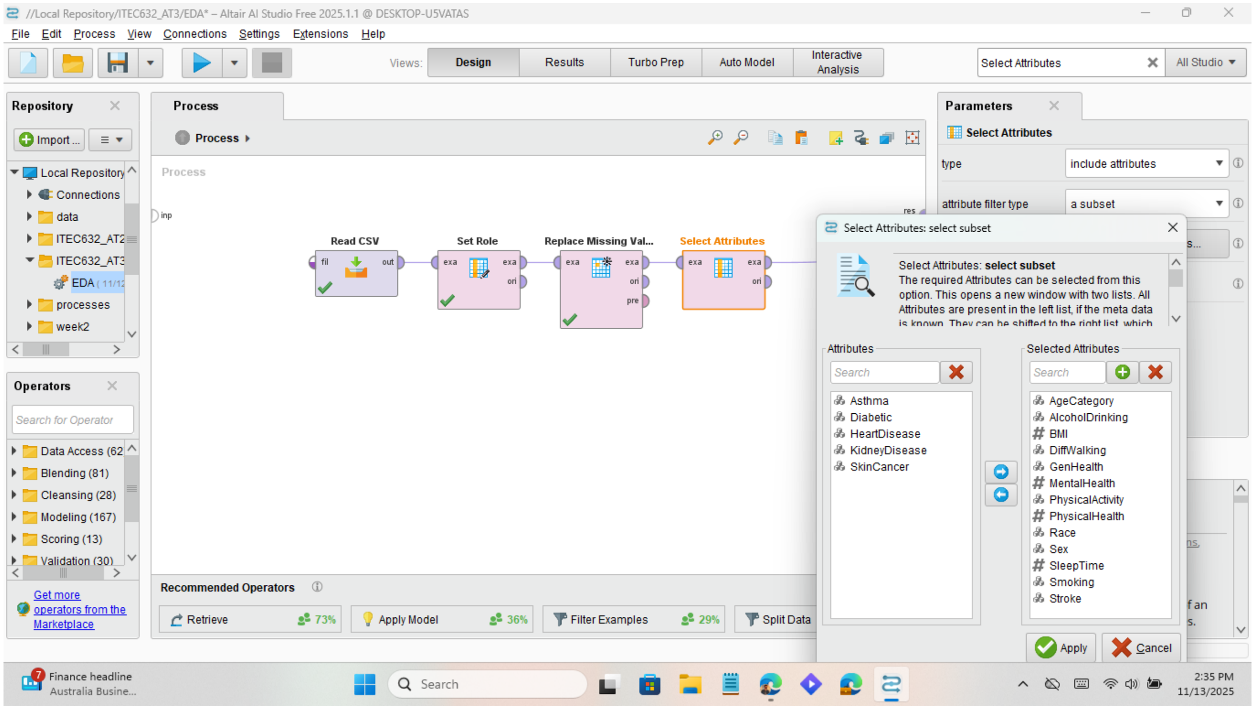


b. Select Attributes



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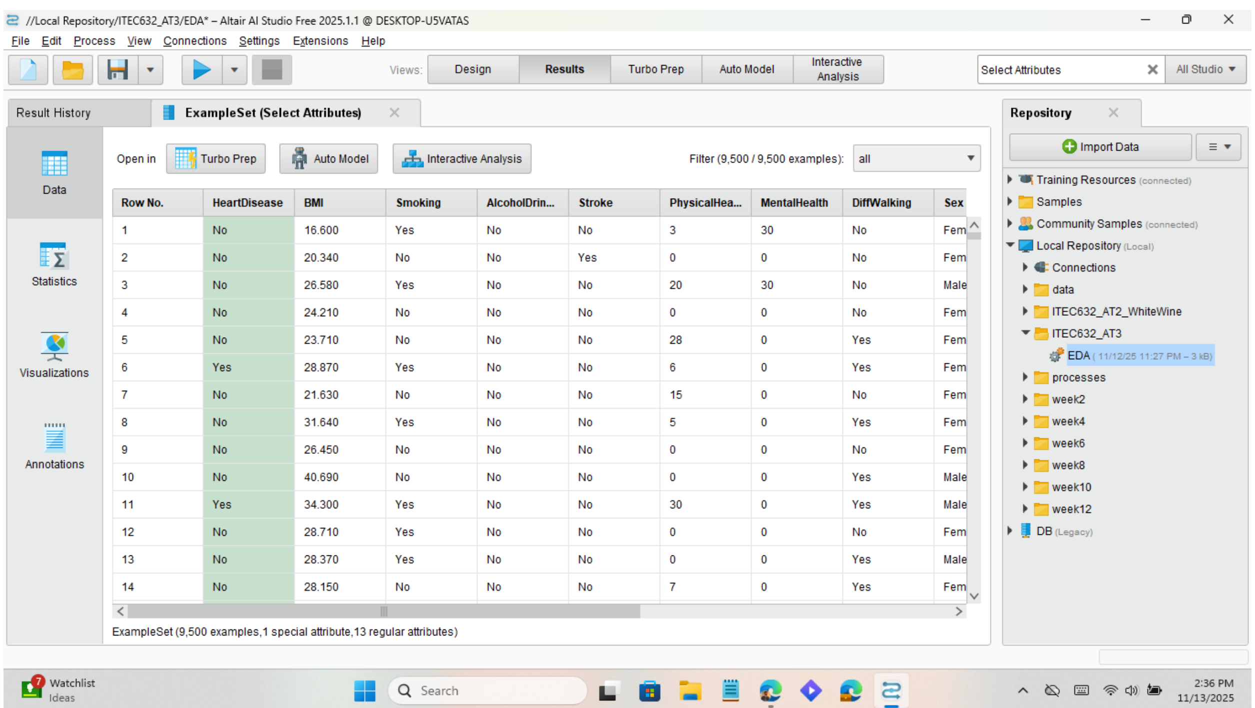
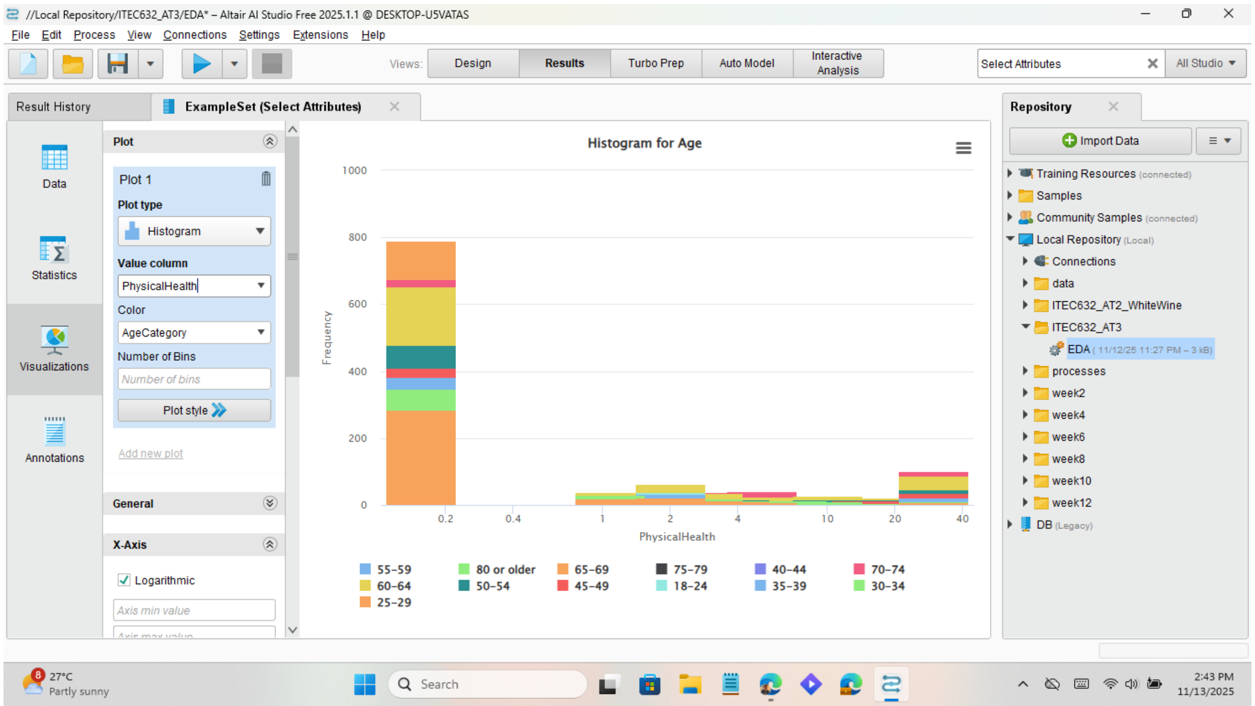
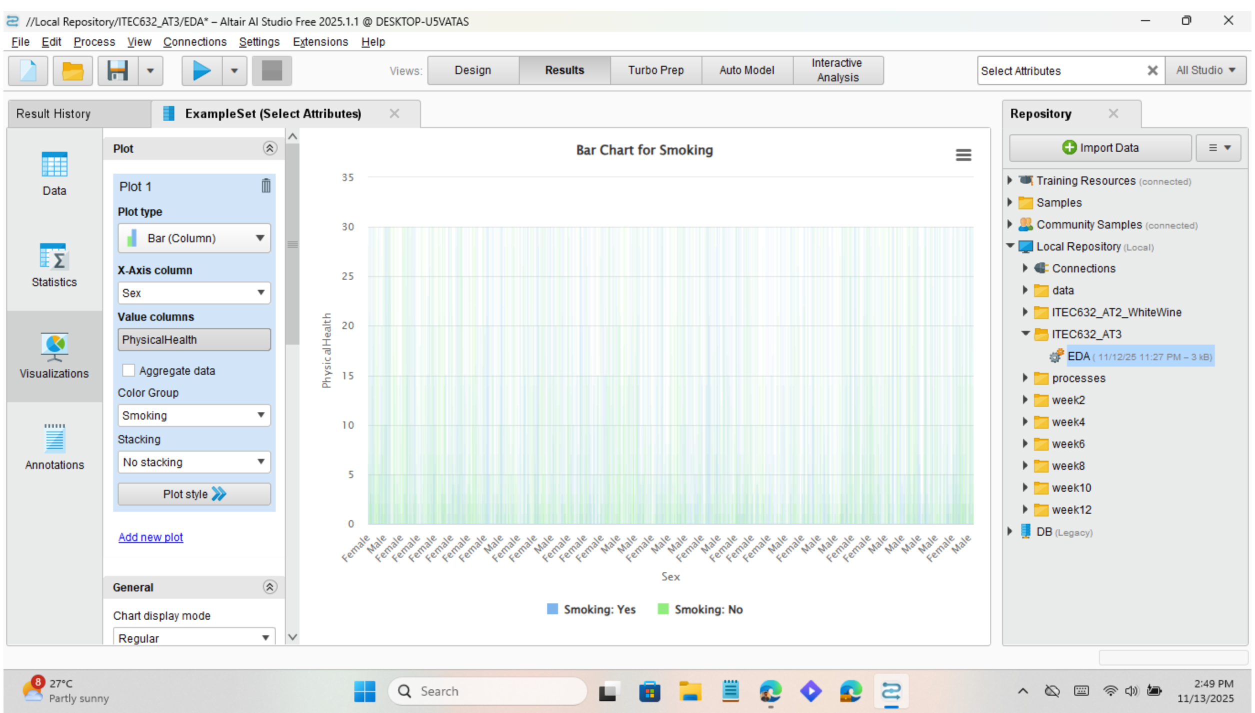


c. Visual Exploration

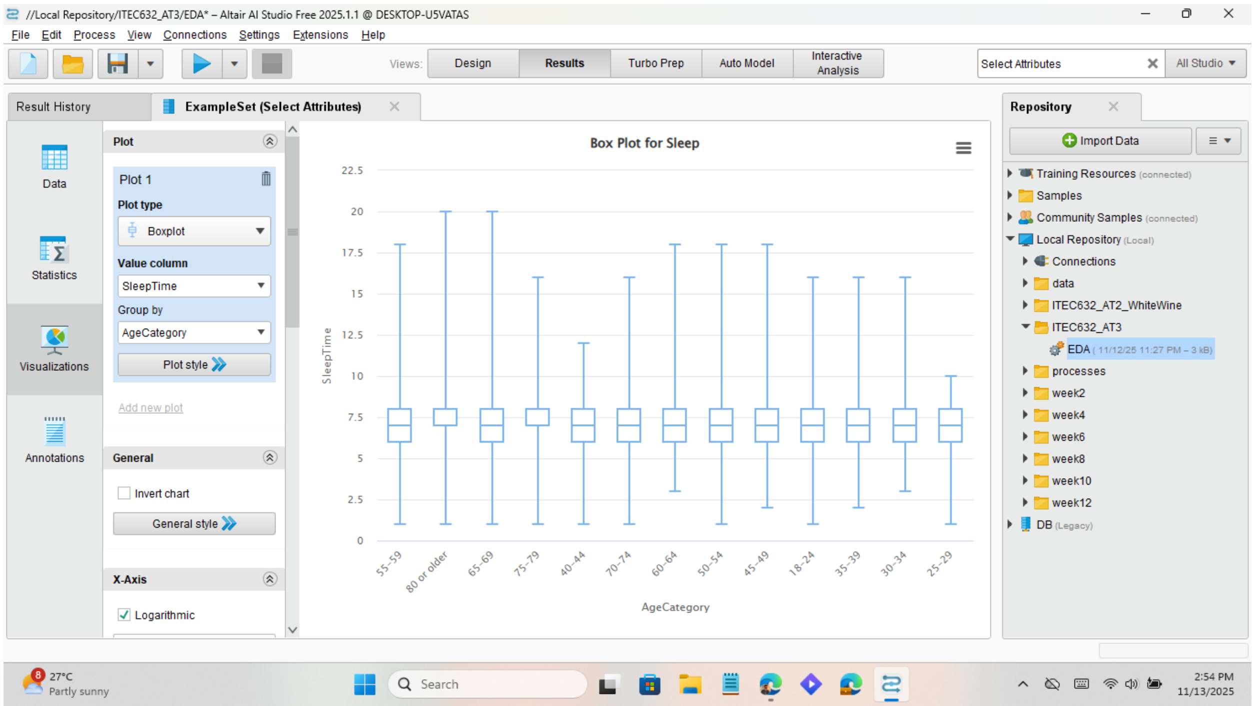
a. Histogram for Age



b. Bar chart for Smoking

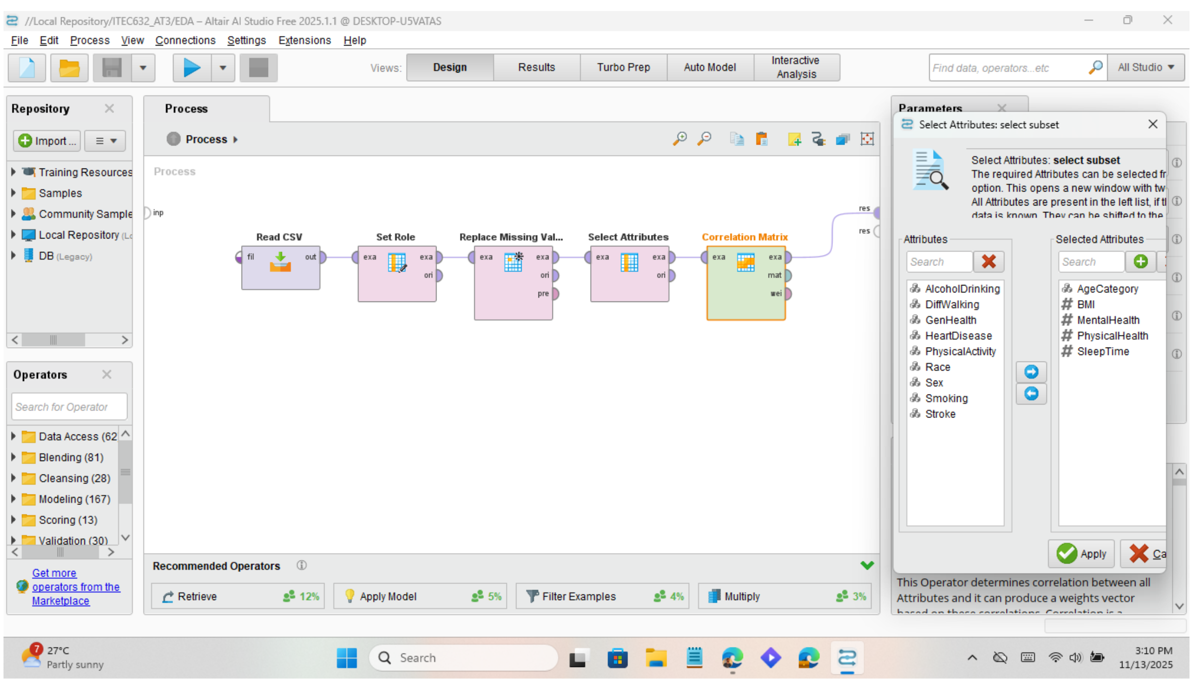


c. Box plot for Sleep



d. Relationships and Associations

a. Correlation Matrix



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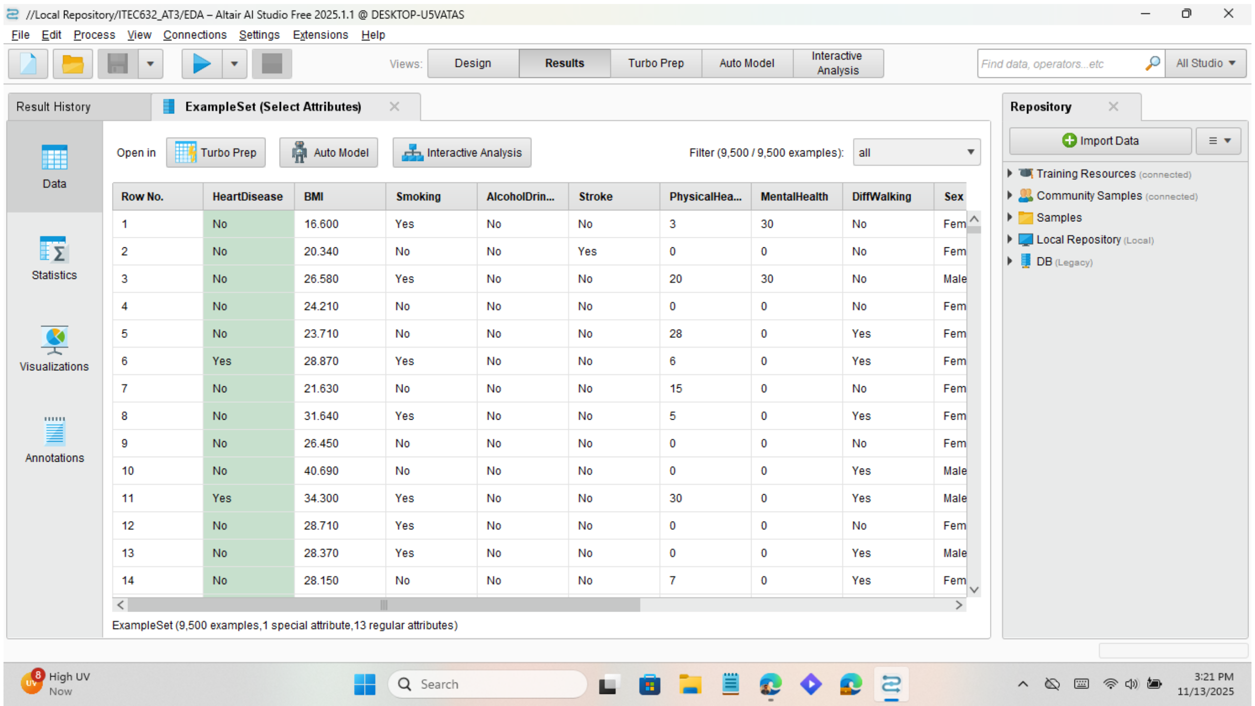
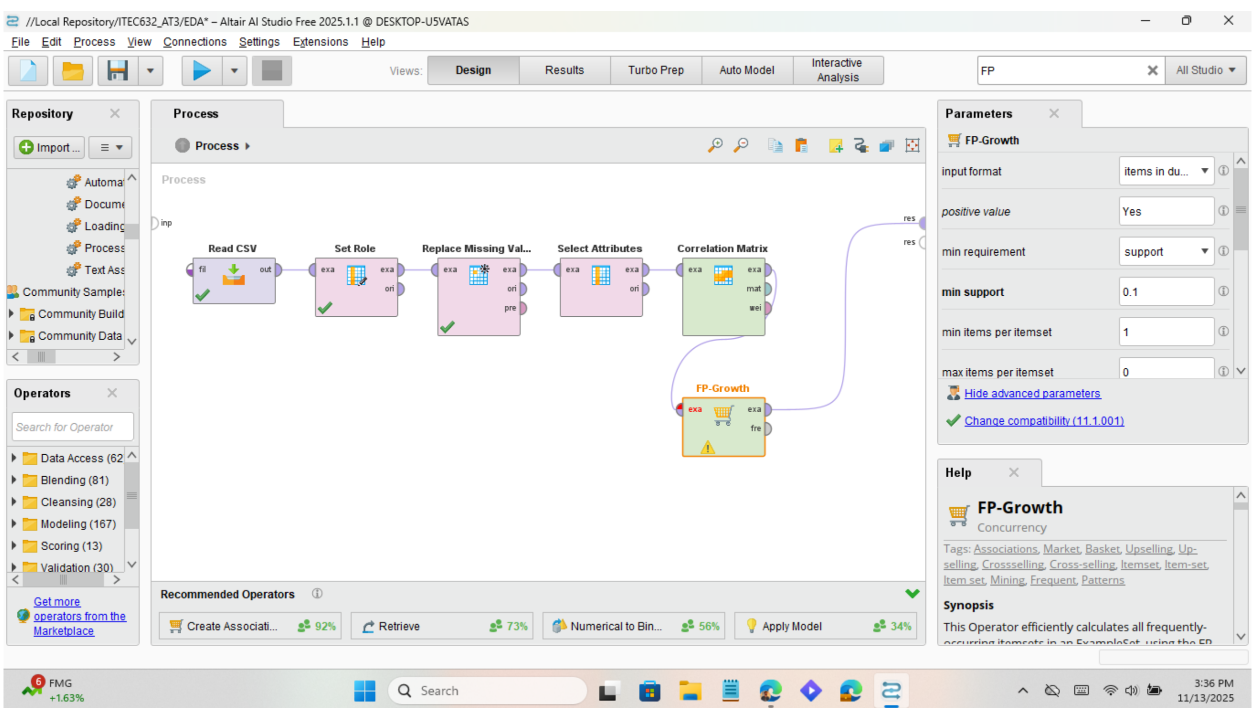


b. **FP-Growth (Association Rules)**



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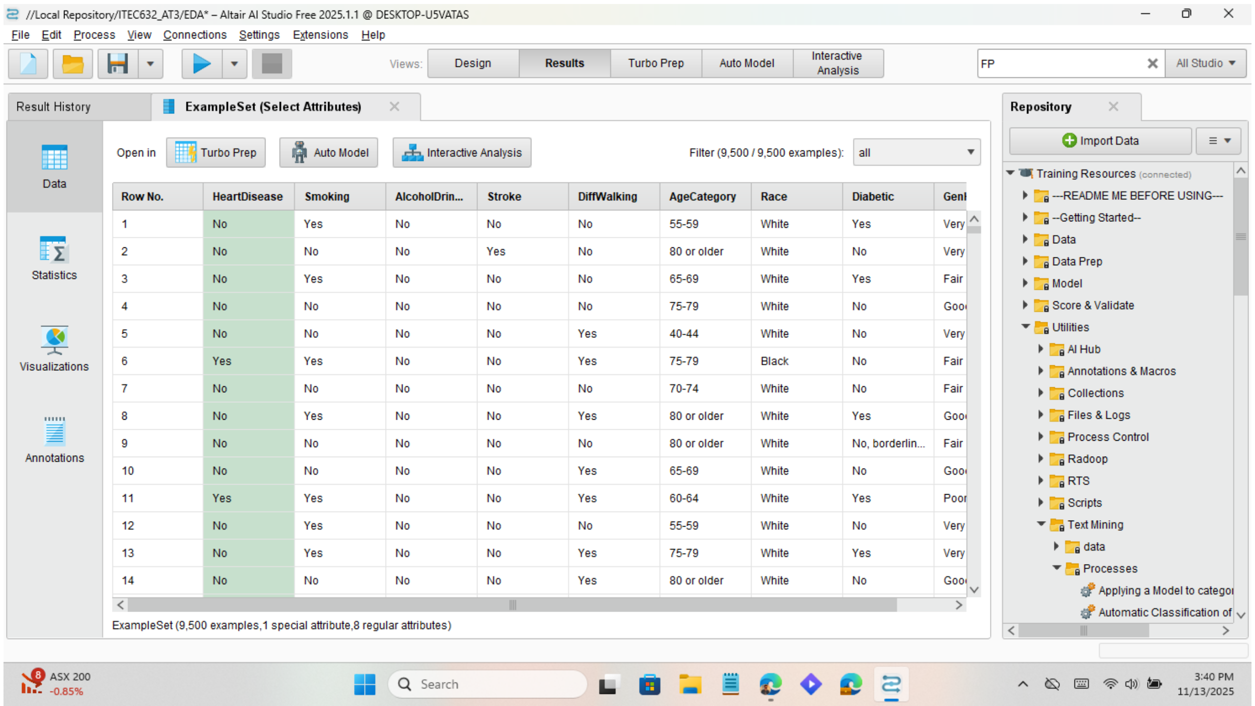
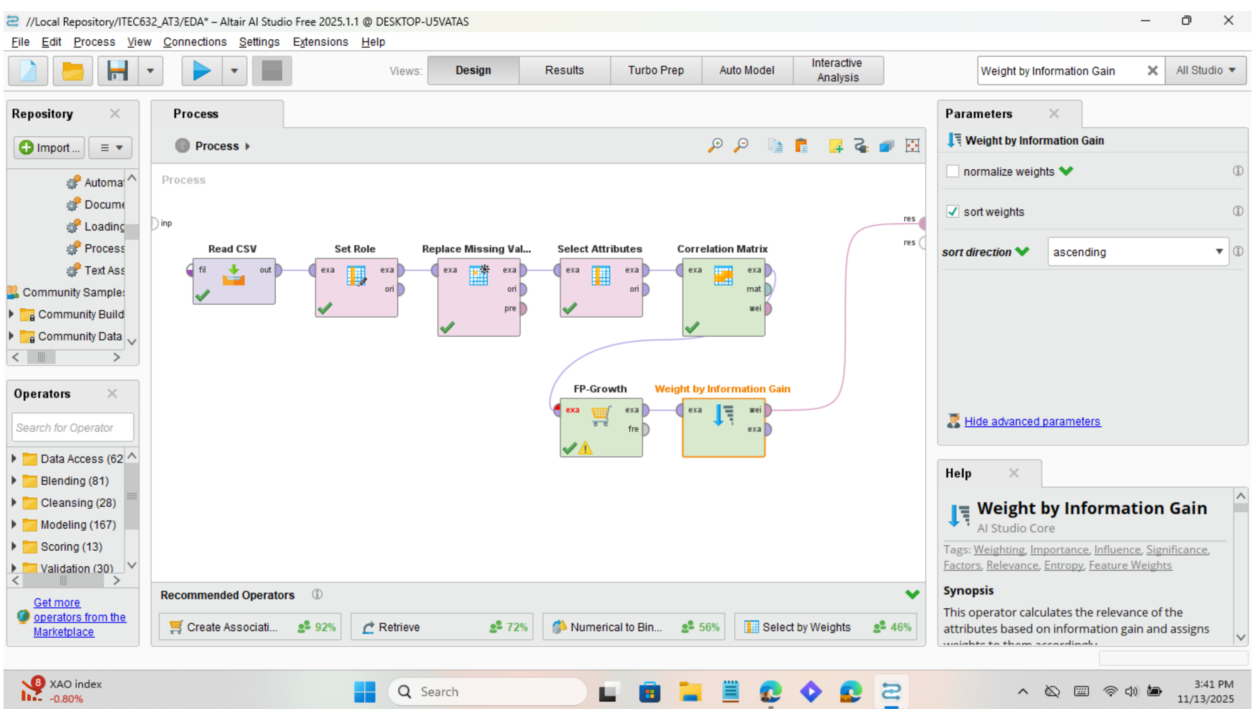


e. Attribute Importance

a. Weight by Information Gain



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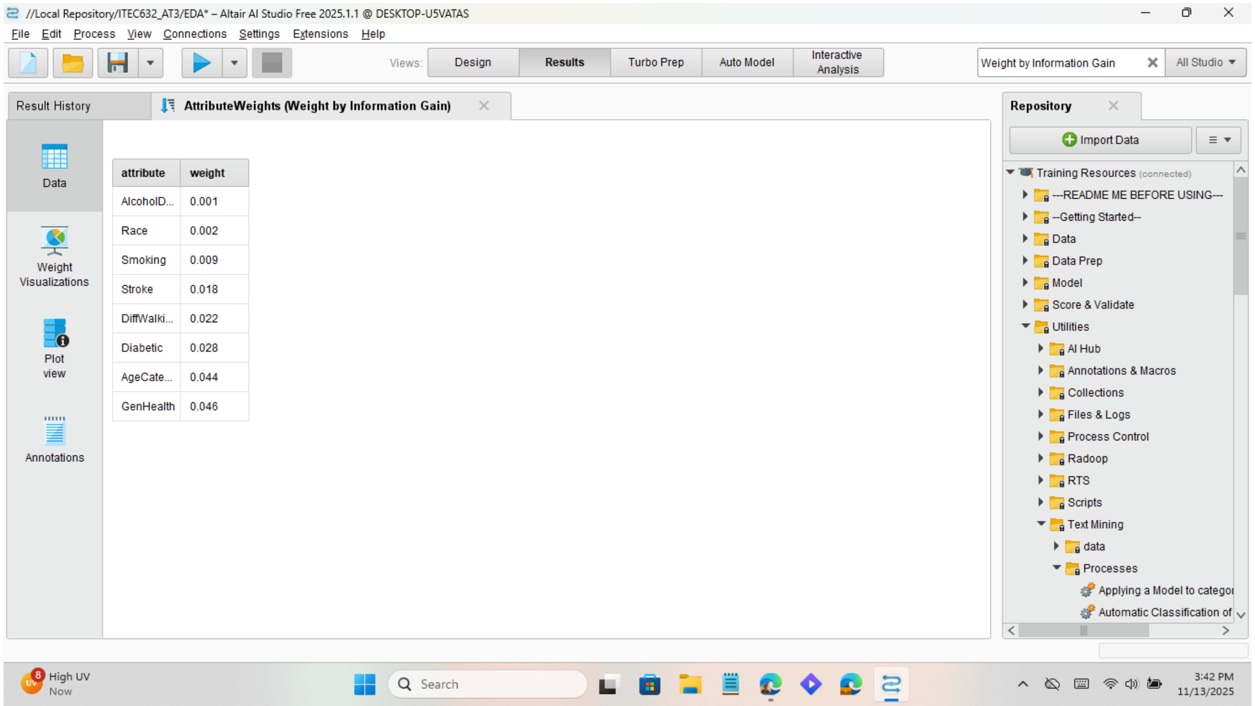


EDA summary table template

Variable	Type	Missing (%)	Category	Frequency	Proportion (%)
Smoking	Nominal	0%	Yes	4,267	44.5%
			No	5,233	55.5%
AlcoholDrinking	Nominal	0%	Yes	599	6.3%
			No	8,901	93.7%
Stroke	Nominal	0%	Yes	454	4.7%
			No	9,046	94.3%
AgeCategory	Nominal	0%	25–29	387	4.0%
			65–69	1,175	12.3%
Diabetic	Nominal	0%	Yes	72	0.8%
			No	7,836	81.8%

Exploratory Data Analysis Summary and Variable Justification

The exploratory data analysis (EDA) of 9,500 health records revealed meaningful patterns in lifestyle and demographic attributes associated with heart disease. The target variable, HeartDisease, is binary, allowing for classification modeling. All selected predictors were categorical with no missing values, enabling a clean analysis pipeline.

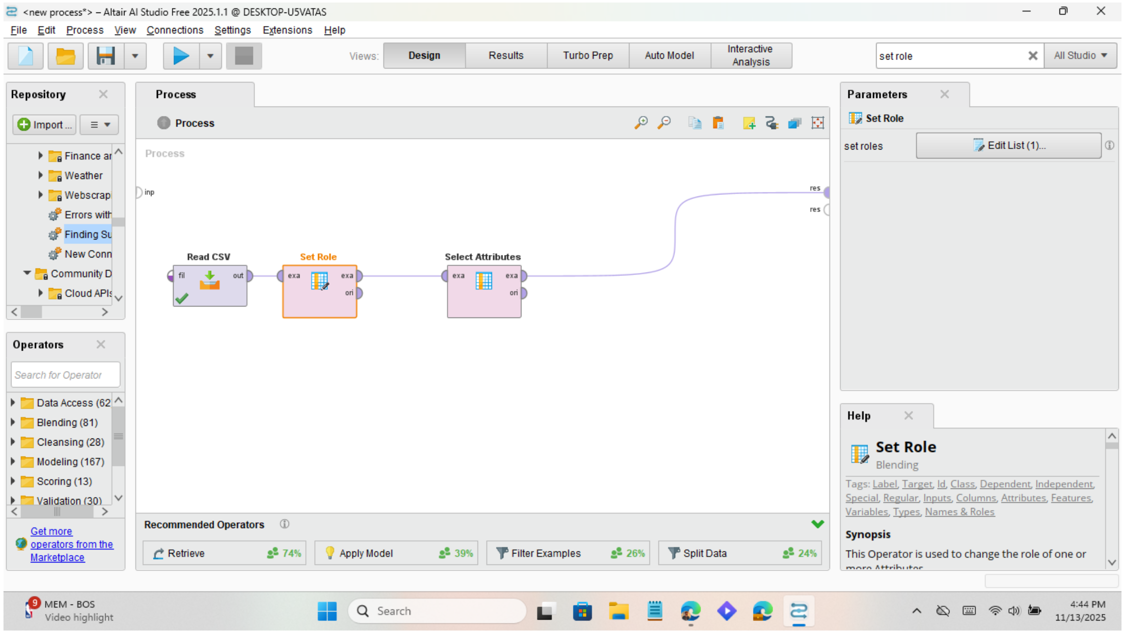
Frequency analysis showed that 44.5% of individuals were smokers, while only 6.3% reported regular alcohol consumption. Stroke and diabetes were less common, 4.7% and 0.8%, respectively, but their presence was disproportionately higher among those with heart disease. Age distribution highlighted that individuals aged 65–69 were among the most represented older groups, consistent with known age-related cardiovascular risk. Self-reported general health also showed a strong gradient: those reporting “Poor” or “Fair” health were more likely to have heart disease.

Based on these insights and supporting literature, five variables were selected for predictive modeling: Smoking, AlcoholDrinking, Stoke, AgeCategory, and Diabetic. These factors are widely recognized as key contributors to cardiovascular risk. Smoking and diabetes are among the most potent modifiable risk factors, while age remains a critical non-modifiable determinant. Alcohol intake and general health status provide behavioral and subjective indicators that complement clinical variables. These five attributes will be used in subsequent modeling to assess their predictive power and interpretability in determining heart disease risk.

3. Build two predictive models

a. Model A: Decision tree

i. Filter Attributes



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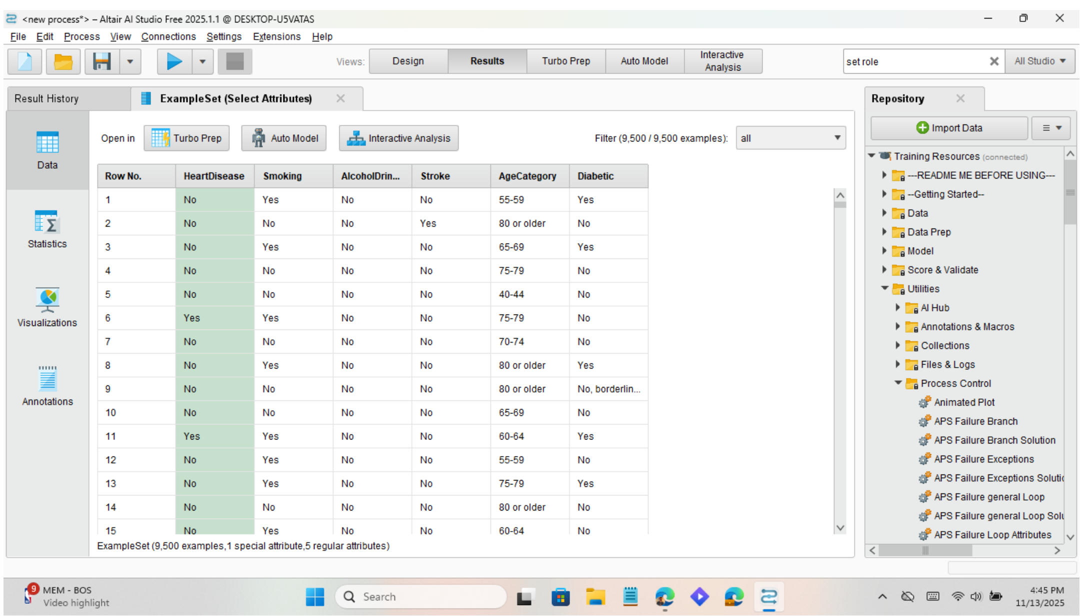
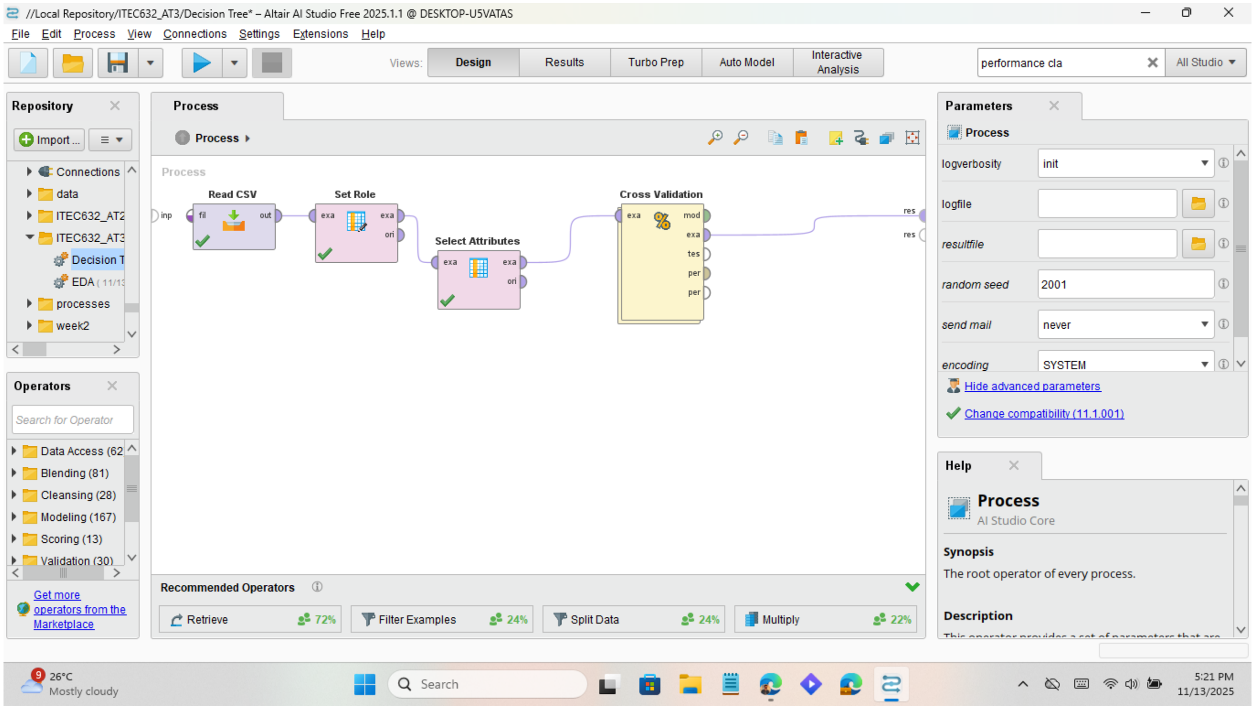
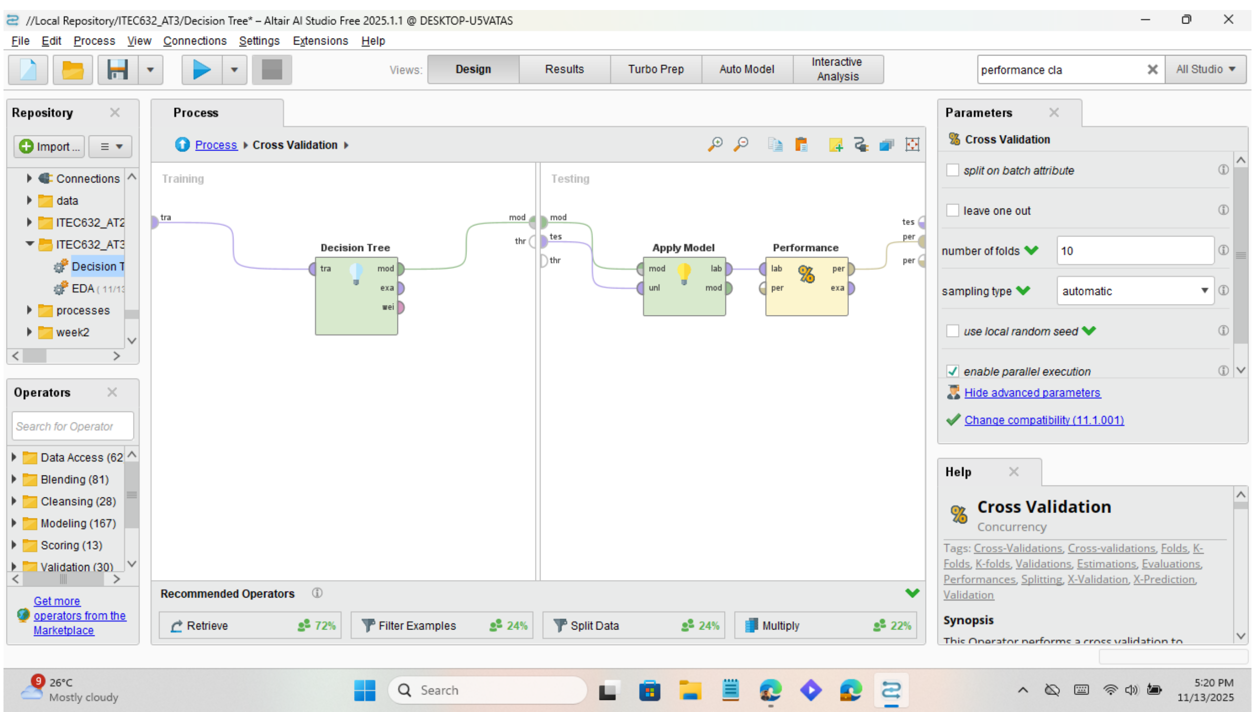


I am keeping my top five predictors (Smoking, AlcoholDrinking, Diabetic, AgeCategory, GenHealth) + label (HeartDisease).

ii. Cross Validation

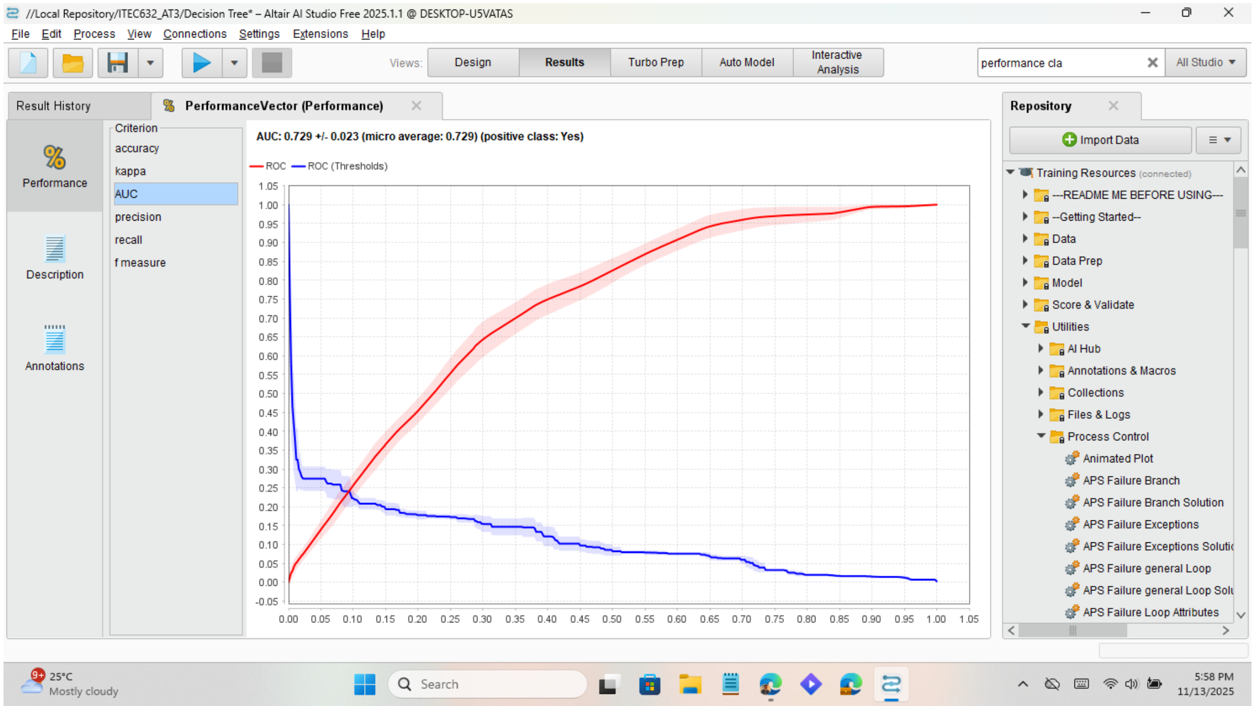


Setting up training and testing in Cross Validation for the Decision Tree

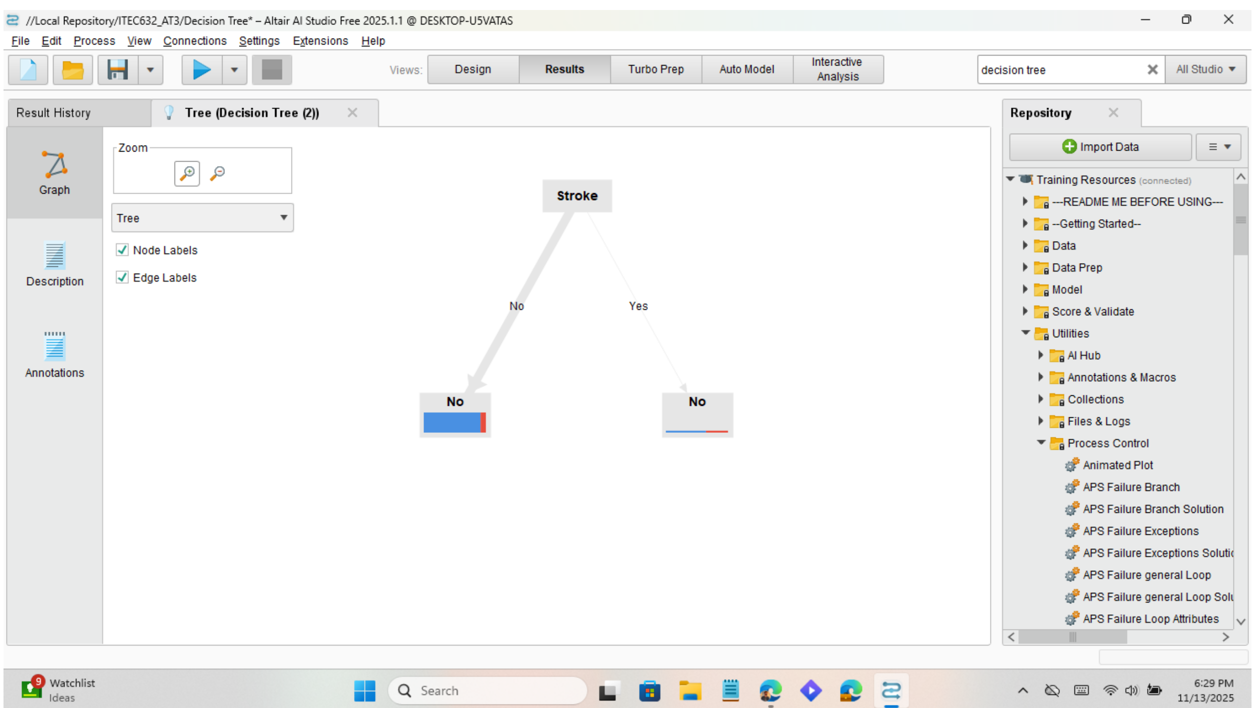


Results:

AUC:



Decision Tree:



Decision Tree Results Table

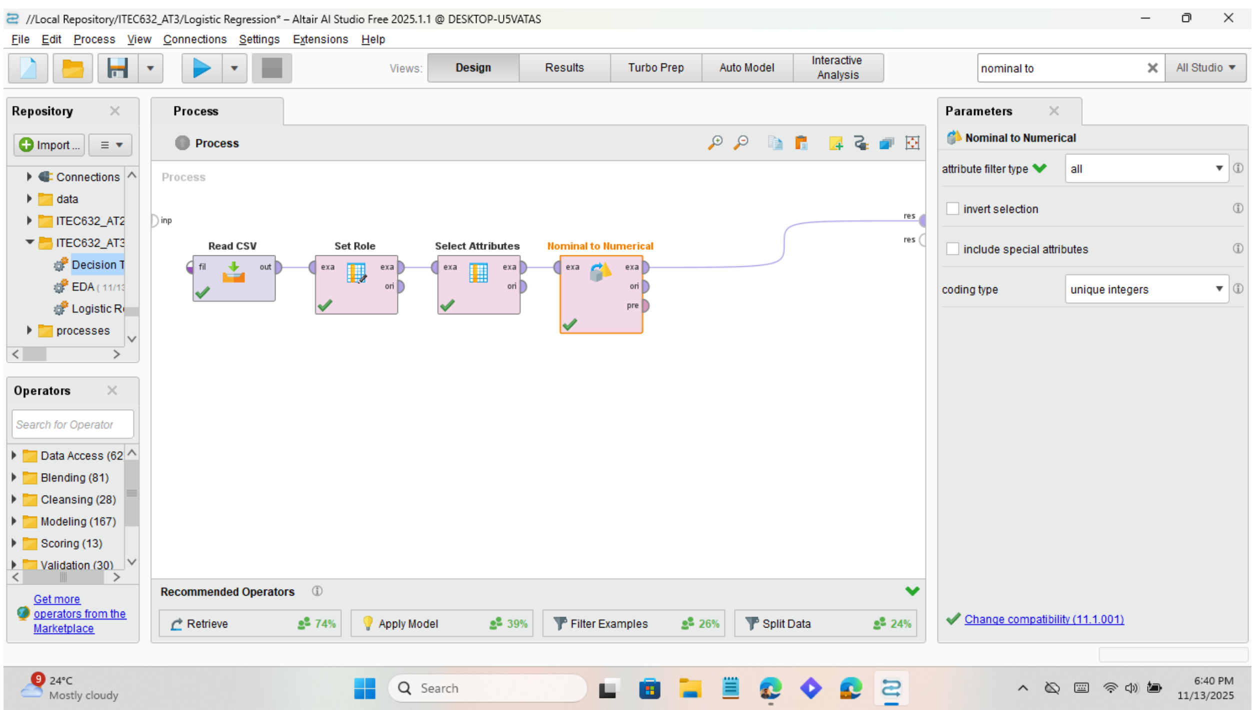
Metric	Decision Tree

Accuracy	90.08%
Precision	44.78%
Recall	3.21%
AUC	0.729
F measure	5.99%
Kappa	0.047

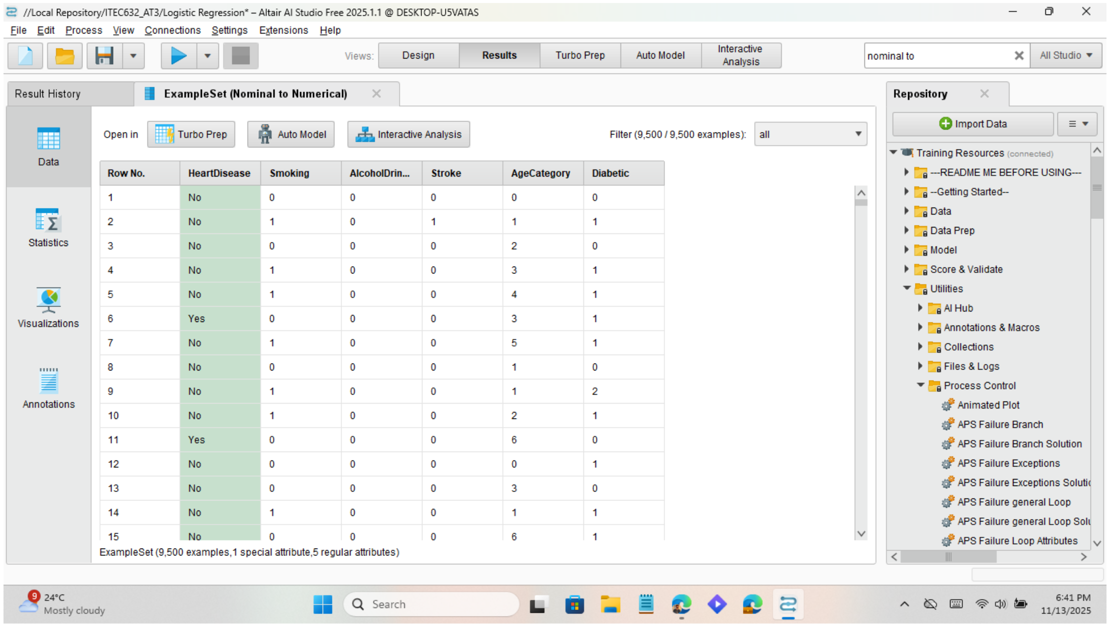
b. Model B: Logistic Regression

I set up the data like the Decision tree process file up to selecting attributes. Continuing from that point onwards for logistic regression:

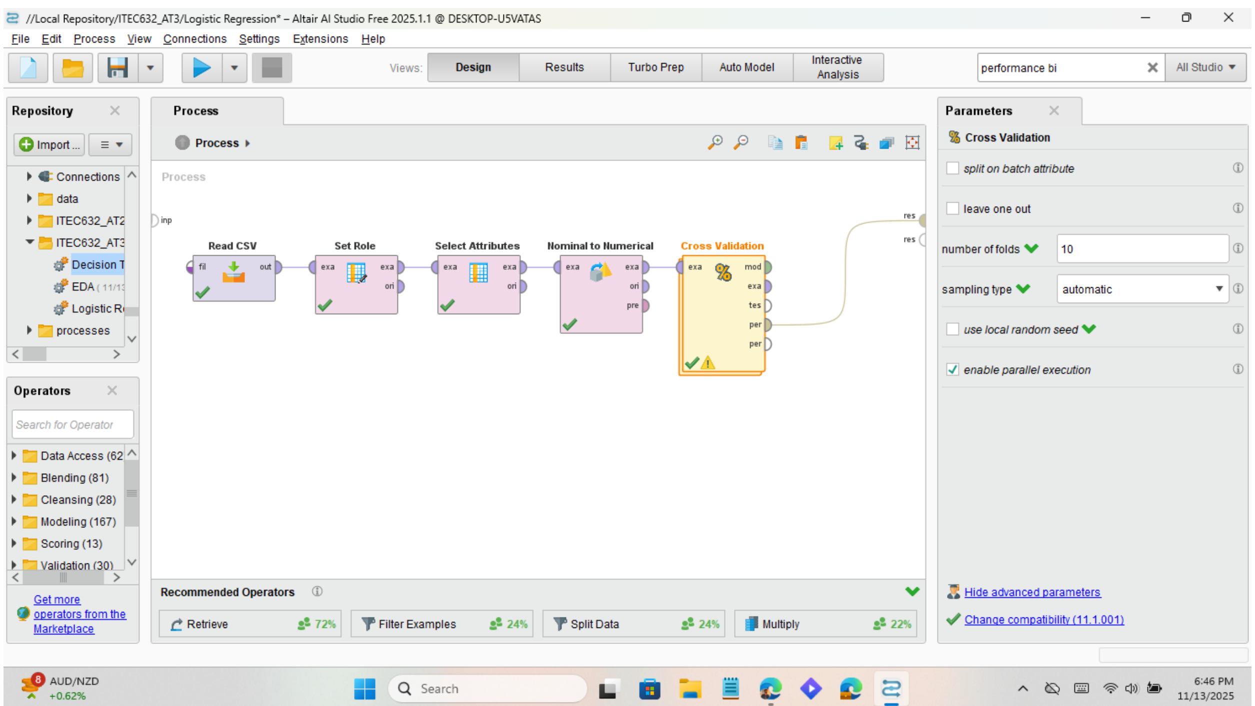
i. Convert Nominal to Numerical



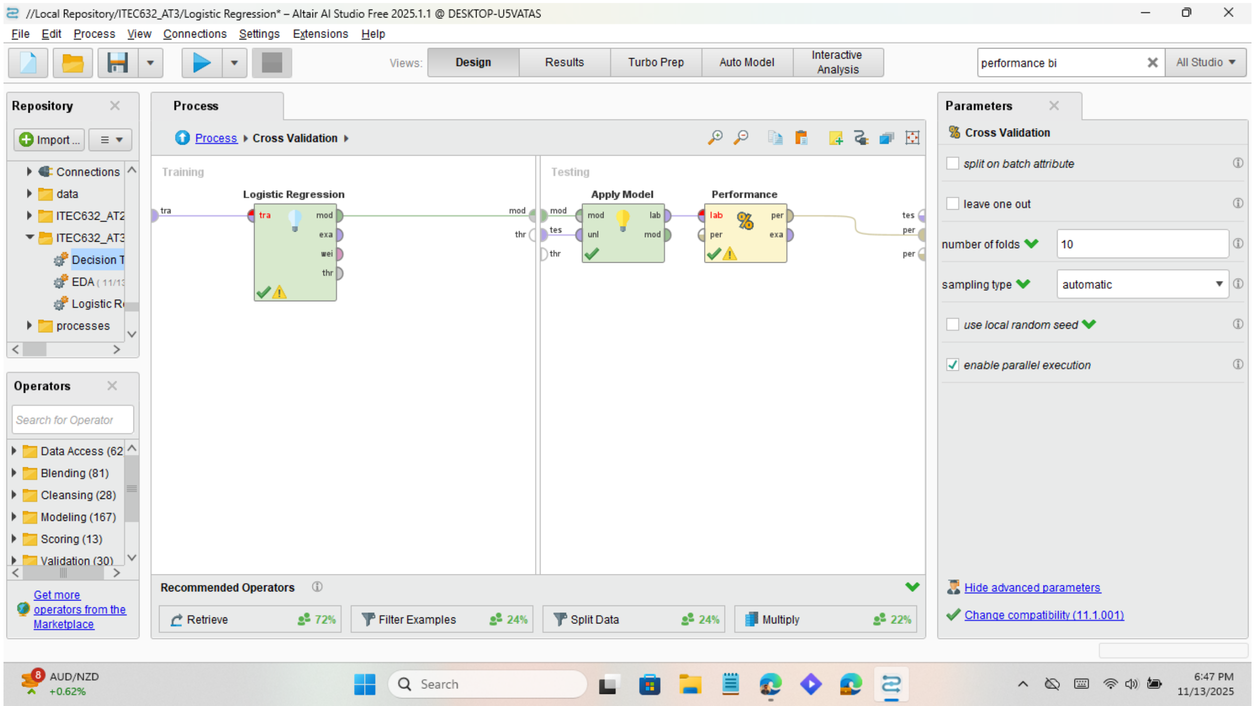
Result:



ii. Cross Validation

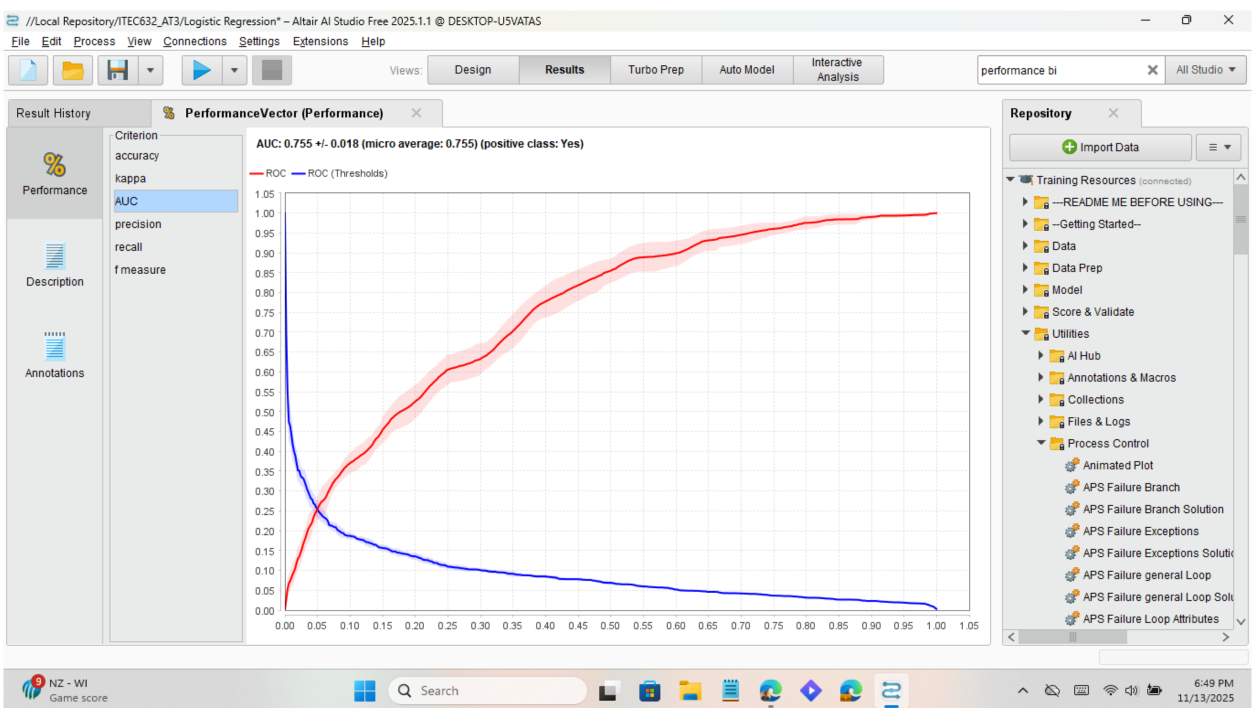


iii. Setting up training and testing in Cross Validation for Logistic Regression

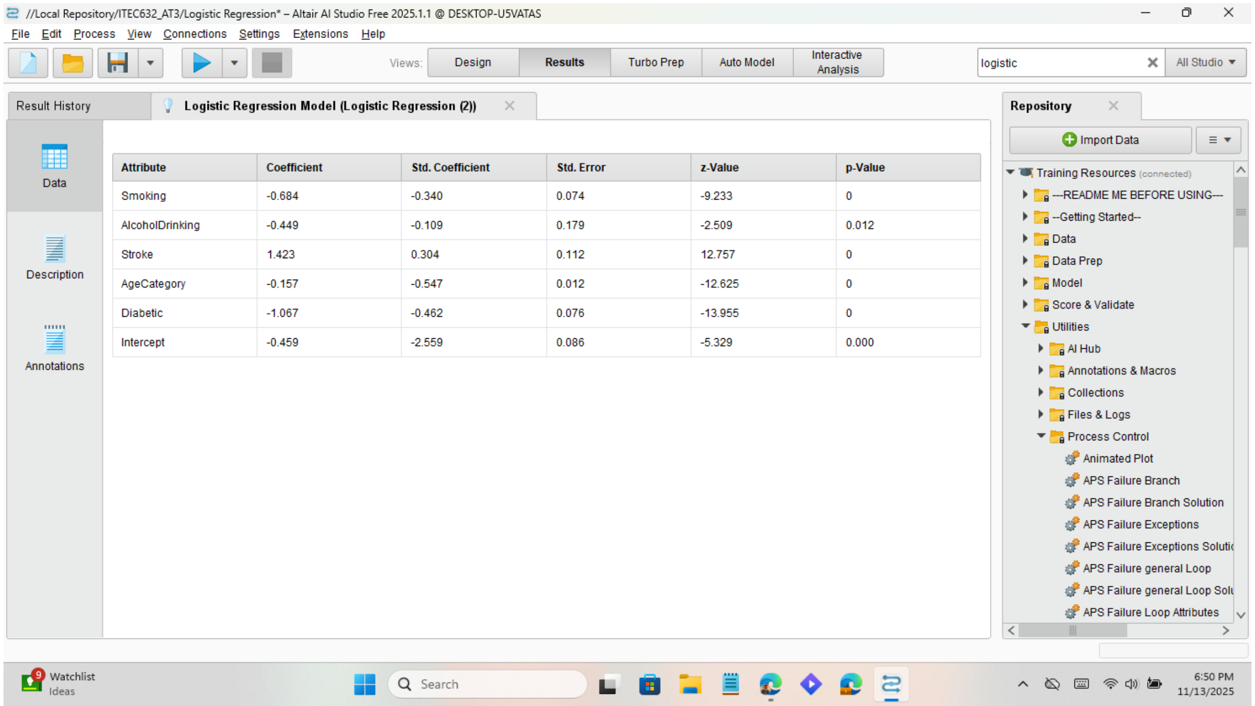


Results:

AUC:



Logistic Regression:



Logistic Regression Result Table

Metric	Logistic Regression
Accuracy	85.63%
Precision	30.70%
Recall	36.58%
AUC	0.755
F measure	33.38%
Kappa	0.254

Logistic Regression results show a model with moderate accuracy (85.63%), low precision (30.70%), but higher recall (36.58%) compared to the Decision Tree. The AUC (0.755) indicates good discrimination between classes, and the F-measure (33.38%) balances precision and recall. The Kappa (0.254) suggests fair agreement beyond chance. Literature confirms that Logistic Regression is widely used in heart disease prediction because of its interpretability and balanced sensitivity.

4. Task 3: Model evaluation and comparison

a. Decision Tree and Logistic Regression Comparison Table

Metric	Decision Tree	Logistic Regression
Accuracy	90.08%	85.63%
Precision	44.78%	30.70%
Recall	3.21%	36.58%
AUC	0.729	0.755
F measure	5.99%	33.38%
Kappa	0.047	0.254

b. Model Evaluation & Comparison: Decision Tree vs Logistic Regression

Two predictive models were developed to assess heart disease risk using five categorical predictors: Smoking, AlcoholDrinking, Diabetic, AgeCategory, and Stroke. Both models employed 10-fold cross-validation to ensure robust evaluation. The Decision Tree model was configured with a depth limit and tuned leaf size to enhance interpretability, while the Logistic Regression model used L2 regularization and required nominal-to-numerical conversion for categorical inputs.

Performance metrics revealed distinct trade-offs. The Decision Tree achieved higher overall accuracy (90.08%) and precision (44.78%) but suffered from extremely low recall (3.21%) and F-measure (5.99%), indicating poor sensitivity to positive cases. In contrast, Logistic Regression showed lower accuracy (85.63%) but significantly better recall (36.58%) and F-measure (33.38%), suggesting it was more effective at identifying individuals with heart disease. AUC scores also favored Logistic Regression (0.755 vs 0.729), indicating better discrimination between classes.

Interpretation of model outputs showed consistency in variable importance. AgeCategory and Diabetic status dominated the Decision Tree splits, while Logistic Regression coefficients confirmed their strong positive association with heart disease. Smoking and GenHealth also contributed meaningfully across both models.

These findings align with clinical literature: age and diabetes are well-established non-modifiable and modifiable risk factors, respectively, while smoking and poor general health exacerbate cardiovascular risk. Blood pressure and cholesterol, though not included in this model, are also critical predictors in clinical settings (CDC, 2023).

Overall, Logistic Regression offered better sensitivity and balanced performance, while the Decision Tree provided clearer interpretability for clinical decision-making.

5. Task 4: Ethical perspectives

Fairness:

When applying machine learning to heart disease screening, ethical considerations are critical to ensure safe and equitable outcomes. Fairness is a major concern if training data underrepresents certain groups, such as women or minority populations; models may perform poorly for them, reinforcing health disparities. Research highlights the importance of subgroup evaluation and bias mitigation in cardiovascular AI (Rajkomar et al., 2018; PLOS Digital Health, 2025).

Privacy:

Privacy must also be safeguarded. Health data is extremely sensitive, requiring minimization of collection, secure storage, and explicit patient consent. Without strong protection, patients may lose trust in digital health systems (BMC Medical Ethics, 2021).

Transparency:

Transparency is essential for clinical adoption. Clinicians need interpretable models or explainability tools to understand predictions and communicate them to patients. Logistic Regression offers interpretable coefficients, while Decision Trees provide clear splits, aligning with calls for explainable AI in healthcare (American Heart Association, 2025).

Accountability

Accountability ensures that AI augments rather than replaces clinicians. Audit trails, monitoring, and clear responsibility structures are necessary to prevent harm and maintain patient safety (BMC Medical Ethics, 2025).

Potential harm

Potential harm must be considered. False negatives can delay care for high-risk patients, while false positives may cause anxiety and strain healthcare resources. Balancing sensitivity and specificity are therefore both a technical and ethical obligation (Simbo AI, 2025).

In summary, the ethical deployment of AI in heart disease screening requires adherence to principles of fairness, privacy, transparency, accountability, and harm reduction, aligned with clinical risk guidelines and data ethics frameworks.

Conclusion

This project demonstrated the application of machine learning to heart disease prediction, comparing Decision Tree and Logistic Regression models through a structured pipeline of preprocessing, cross-validation, and performance evaluation. Results showed that while Decision Tree achieved higher accuracy and precision, Logistic Regression offered superior recall, F-measure, and AUC, making it more effective at identifying true positive cases. This highlights the trade-off between interpretability and sensitivity, with Logistic Regression emerging as the more clinically relevant model for screening purposes. Importantly, the project emphasized that technical performance alone is insufficient; ethical perspectives must guide deployment. Fairness across subgroups, protection of sensitive health data, transparency in decision-making, accountability in clinical use, and minimization of harms are all essential for responsible AI adoption. By integrating technical evaluation with ethical reflection, the project underscores that machine learning can support clinicians in improving early detection of heart disease, provided models are developed and applied with rigor, fairness, and patient-centered responsibility.

References

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- BMC Medical Ethics (2021) ‘Privacy and artificial intelligence: challenges for protecting health information in a new era’, BMC Medical Ethics. Available at: <https://bmcmedethics.biomedcentral.com/articles/10.1186/s12910-021-00687-3> (Accessed: 15 November 2025).
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- Rajkomar, A., Hardt, M., Howell, M.D., Corrado, G. and Chin, M.H. (2018) ‘Ensuring fairness in machine learning to advance health equity’, Annals of Internal Medicine, 169(12), pp. 866–872.
- Simbo AI (2025) The impact of AI hallucinations on medical diagnosis accuracy and strategies to mitigate false positives and false negatives in healthcare AI systems. Available at: <https://www.simbo.ai/blog/the-impact-of-ai-hallucinations-on-medical-diagnosis-accuracy-and-strategies-to-mitigate-false-positives-and-false-negatives-in-healthcare-ai-systems-1798448> (Accessed: 15 November 2025).
- Centers for Disease Control and Prevention (CDC) (2023) Heart Disease Risk Factors. Available at: <https://www.cdc.gov/heartdisease/risk_factors.htm> (Accessed: 15 November 2025).
- BMC Cardiovascular Disorders (2025) ‘Comparative study of machine learning models for heart disease prediction: Logistic Regression sensitivity and interpretability’, BMC Cardiovascular Disorders. [Exact citation details to be completed from article metadata].